# §9. Parametric Eqn & Polar Coord.

## · Parametric Equation

- **Definition**

  : For a parameter $t$, an equation $\begin{cases} x = f(t), \\ y = g(t). \end{cases}$

- **Eliminating the Parameter**

  : Changing $\begin{cases} x = f(t), \\ y = g(t) \end{cases}$ to $F(x, y) = 0$.

  But by this progress, the direction disappears.

- **An Example of P.Eqn**

  : A circle s.t. its center is on O, and its radius is $1$ can be written to

  $$x = \cos t, \quad y = \sin t, \quad 0 \leqslant t \leqslant 2\pi.$$

## · Cycloid

- **Definition**

  : A parametric eqn. family

  $$\begin{cases} x = r(r - \sin\theta), \\ y = r(1 - \cos\theta) \end{cases}$$

  with the parameter $\theta \in \mathbb{R}$ and the const. $r \in \mathbb{R}^+$.

- the fastest landing curve. (=brachistochrome)

## · Para.Eqn. Family

- **Example**: For $a \in \mathbb{R}$, $\begin{cases} x = a + \cos t, \\ y = a\tan t + \sin t. \end{cases}$
  - $|a| > 0$: asymptote $(x = a)$.
  - $|a| = 1$: $\exists$ 'cusp' at O.
  - $a = 0$: Circle s.t. center $=$ O and $r = 1$.

## · Calculus of Para.Eqn.

- **Diff.: Tangent** ($y$: d.able fcn. of $x$)
  - $\dfrac{\mathrm{d}x}{\mathrm{d}t} \neq 0$; $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$.
  - $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 0$; vertical asym. at $t$ s.t. $x'(t) = 0$.
  - $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 0$; horizontal tangent at $t$ s.t. $y'(t) = 0$.

pf.

Firstly, by definition, we know that

$$\frac{\mathrm{d}y}{\mathrm{d}x} \equiv \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(t + \Delta t) - y(t)}{x(t + \Delta t) - x(t)}.$$

By MVT, $\exists t_x^*, t_y^* \in (t, t + \Delta t)$ s.t.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{\Delta x \to 0} \frac{\Delta t \cdot y'(t_y^*)}{\Delta t \cdot x'(t_x^*)}.$$

Then $\Delta t = \dfrac{\Delta x}{x'(t_x^*)} \longrightarrow 0$ as $\Delta x \longrightarrow 0$ by the assertion $x'(s) \neq 0\ \forall s$, hence $t_x^*, t_y^* \longrightarrow t$.
Thus we have, as a result,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \lim_{t_x^*, t_y^* \to t} \frac{\cancel{\Delta t} \cdot y'(t_y^*)}{\cancel{\Delta t} \cdot x'(t_x^*)} = \frac{y'(t)}{x'(t)} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}.$$

■

- **Int.: Length**

  : A length of a curve $S$ s.t. $(x(t), y(t))$ $(\alpha \leqslant t \leqslant \beta)$ is

  $$L = \int_\alpha^\beta \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t.$$

pf.

For $n \in \mathbb{N}$, $\{\alpha, \beta\} \subset \mathrm{dom}(S)$ $(\alpha \leqslant \beta)$, let $\Delta t := (\beta - \alpha)/n$, $\langle t_0, t_n\rangle := \langle \alpha, \beta\rangle$, $t_{i+1} = t_i + \Delta t$, $\Delta f_{i+1} = f(t_{i+1}) - f(t_i)$ for $f_i \in \{x_i, y_i\}$, $\forall i \in [0, n) \cap \mathbb{Z}$.

For $\Delta L_i := \dfrac{\Delta y_i}{\Delta x_i}$, by MVT, $\exists t_{i,x}^*, t_{i,y}^* \in (t_i, t_{i+1})$ s.t.

$$\begin{aligned} \Delta L_i &= \sqrt{\left[\Delta t \cdot \frac{\mathrm{d}}{\mathrm{d}t} x(t_{i,x}^*)\right]^2 + \left[\Delta t \cdot \frac{\mathrm{d}}{\mathrm{d}t} y(t_{i,y}^*)\right]^2} \\ &= \Delta t \sqrt{\left[\frac{\mathrm{d}}{\mathrm{d}t} x(t_{i,x}^*)\right]^2 + \left[\frac{\mathrm{d}}{\mathrm{d}t} y(t_{i,y}^*)\right]^2} \end{aligned}$$

Since $\Delta t = (\beta - \alpha)/n \longrightarrow 0^+$ as $n \longrightarrow \infty$, and $t_{i,x}^*, t_{i,y}^* \longrightarrow t_i$ as $\Delta t \longrightarrow 0^+$, we have

$$\begin{aligned} L &= \lim_{n\to\infty} \sum_{i=1}^n \Delta L_i \\ &= \lim_{n\to\infty} \sum_{i=1}^n \Delta t \sqrt{\left[\frac{\mathrm{d}}{\mathrm{d}t} x(t_{i,x}^*)\right]^2 + \left[\frac{\mathrm{d}}{\mathrm{d}t} y(t_{i,y}^*)\right]^2} \\ &= \int_\alpha^\beta \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t. \end{aligned}$$

■

- **Int.: Area of 'Solid of Revolution'**

  : The area of surface that revolved the graph of $(f(t),\, g(t))$ $(\alpha \leqslant t \leqslant \beta)$ around $x$-axis:

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2}\,\mathrm{d}t.$$

- **Bézier curve**

  : A curve with 4 *control points* $\mathrm{P}_i(x_i,\, y_i)$, $(i = 0,\, 1,\, 2,\, 3)$

$$x = \sum_{i=0}^{3} x_i (1-t)^{3-i} t^i, \quad y = \sum_{i=0}^{3} y_i (1-t)^{3-i} t^i$$

  with a parameter $0 \leqslant 1 \leqslant t$.

  This is used for the digitalization of fonts, etc.

## · Polar Coordinate

- **Definitions**
  - · Pole: O on Cart.Coord.
  - · Polar Axis: $x^+$-axis on Cart.Coord.
  - · A point $\mathrm{P}(r,\, \theta)$;
    - $r \equiv \overline{\mathrm{OP}}$,
    - $\theta = \angle \mathrm{XOP}$ for $\mathrm{X}(x,\, 0)$ $(x \in \mathbb{R}^+)$.
  - · $(-r,\, \theta) = (r,\, \pi + \theta)$

- **Symmetry** (for a polar equation $F\colon f(r,\, \theta) = 0$)
  - · $f(-r,\, \theta) = f(r,\, \theta) \Longrightarrow$ p.axis symmetry

- **Area & Length**
  - · $A = \int_a^b \frac{1}{2}[f(\theta)]^2\,\mathrm{d}\theta.$

pf.

For an area of the sector $r^2\theta/2$, and by MVT, we have

$$\Delta A_i = \frac{1}{2}[f(\theta_i^*)]^2 \Delta\theta.$$

Thus, since $A \approx \sum \Delta A_i$, we conclude

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{2}[f(\theta_i^*)]^2 \Delta\theta = \int_a^b \frac{1}{2}[f(\theta)]^2\,\mathrm{d}\theta.$$

■

  - · $L = \int_a^b \sqrt{r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2}\,\mathrm{d}\theta.$

pf.

For a length of a segment $\sqrt{(\Delta x)^2 + (\Delta y)^2}$ and $(r,\, \theta)_{\mathrm{Pol}} = (r\cos\theta,\, r\sin\theta)_{\mathrm{Cart}}$, also by MVT, we have

$$\begin{aligned}\Delta L_i &= \sqrt{[x'(\theta_i^*)]^2 + y'[(\theta_i^*)]^2}\,\Delta\theta \\ &= \sqrt{(r'\cos\theta - r\sin\theta)^2 + (r'\sin\theta + r\cos\theta)^2}\Delta\theta \\ &= \sqrt{r^2 + (r')^2}\Delta\theta.\end{aligned}$$

Thus, since $L = \sum \Delta L_i$, we conclude

$$\begin{aligned}L &= \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{r^2 + (r')^2}\Delta\theta \\ &= \int_a^b \sqrt{r + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2}\,\mathrm{d}\theta.\end{aligned}$$

■

- **Tangent**
  - · $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$, if $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$.

## · Conic Sections

- **Parabolas:** $x^2 = 4py$
  : A set of points which are in same distance between a focus(F, esp. $(0,\, p)$) and a directix(esp. $y = -p$).

- **Ellipses:** $x^2/a^2 + y^2/b^2 = 1$
  : A set of points where the sum of distances between the focus($\mathrm{F}_1$ & $\mathrm{F}_2$, esp. $(\pm c,\, 0)$) is same.
  - · $a$: Half of major axis(segment between two vertexes)
  - · $b = \sqrt{a^2 - c^2}$.

- **Hyperbola:** $x^2/a^2 - y^2/b^2 = 1$

: A set of points where the difference of distances between the focus($\mathrm{F}_1$ & $\mathrm{F}_2$, esp. $(\pm c,\, 0)$) is same.

· $a$: Half of the length of a major axis (segment between two vertices)

· $b^2 = c^2 - a^2$.

pf.

For $\mathrm{F}_1(c,\, 0)$ and $\mathrm{F}_2(-c,\, 0)$, let $a$, $c$ be the positive real number.
Then, for a point $\mathrm{P}(x,\, y)$,

$$|\overline{\mathrm{PF}_1}| = \sqrt{(x-c)^2 + y^2},$$
$$|\overline{\mathrm{PF}_2}| = \sqrt{(x+c)^2 + y^2}.$$

Now, consider the equation of $c$ s.t.

$$|\overline{\mathrm{PF}_1}| - |\overline{\mathrm{PF}_2}| = \pm 2a \iff |\overline{\mathrm{PF}_1}| = |\overline{\mathrm{PF}_2}| \pm 2a,$$

which may be the equation of a hyperbola.
Then we compute

$$\sqrt{(x-c)^2 + y^2} = \sqrt{(x+c)^2 + y^2} \pm 2a;$$
$$x^2 - 2cx + y^2$$
$$= x^2 + 2cx + y^2 \pm 4a\sqrt{(x+c)^2 + y^2} + 4a^2;$$
$$4cx + 4a^2 = \mp 4a\sqrt{(x+c)^2 + y^2};$$
$$cx + a^2 = \mp a\sqrt{(x+c)^2 + y^2};$$
$$c^2x^2 + \cancel{2a^2cx} + a^4 = a^2x^2 + \cancel{2a^2cx} + a^2c^2 + a^2y^2;$$
$$(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2);$$
$$bx^2 - a^2y^2 = a^2b^2;$$
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

for $b \in \mathbb{R}$ s.t. $b^2 = c^2 - a^2$, since $c > a$ in case of *hyperbolas*.
If we had computed the equation of *ellipses*, then we would have let $b^2 = a^2 - c^2$, which is $\times(-1)$ to the one of the hyperbolas. ■

# §11. Vec.'s & Geometry of Space

## · 3D Cartesian Coord. Space

- **Projection onto a plane**

: $\mathrm{P}'$ is a projection onto a plane $L$ of $\mathrm{P}$, if $\mathrm{P}'$ is a foot of perpendicular of $\mathrm{P}$ onto $L$.

ex) $(2,\, 3,\, 0)$ is a projection onto $xy$-plane of $(2,\, 3,\, 1)$.

## · Vectors in $\mathbb{R}^3$

- **Definition**

: $\vec{v} = \mathbf{v} \equiv \langle v_1,\, v_2,\, v_3 \rangle \in \mathbb{R}$.

- **Plain with 2 Vectors**

: $\{c\mathbf{u} + d\mathbf{v} : c,\, d \in \mathbb{R}\}$ for vectors $\mathbf{u}$, $\mathbf{v} \in V \setminus \{\mathbf{0}\}$ represents a plane $L$ including $\mathbf{u}$, $\mathbf{v}$.

*Finding* $\mathbf{N}$ *s.t.* $\mathbf{N} \perp L$ *is a goal of §11.*

- **Vector Spaces** *(From Ln.Algebra)*

: A triple $\langle V,\, \oplus,\, \odot \rangle$ is a vector space, if $c\mathbf{u} + d\mathbf{v} \in V$ for all $c,\, d \in \mathbb{R}$ and $\mathbf{u},\, \mathbf{v} \in V$.

## · Operation between Vectors

$(\mathbf{u} = \langle u_1,\, u_2,\, u_3 \rangle,\ \mathbf{v} = \langle v_1,\, v_2,\, v_3 \rangle \in \mathbb{R}^3,\ c \in \mathbb{R})$

- **Vector Addition**

: $\mathbf{u} + \mathbf{v} \equiv \langle u_1 + v_1,\ u_2 + v_2,\ u_3 + v_3 \rangle$.

- **Scalar Multiplication**

: $c\mathbf{v} \equiv \langle cv_1,\, cv_2,\, cv_3 \rangle$.

- **Inner Product / Dot Product**

: $\mathbf{u} \cdot \mathbf{v} \equiv \sum_{i=1}^{3} u_i v_i$.

- **Outer Product / Cross Product**

$$\mathbf{u} \times \mathbf{v} \equiv \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$
$$= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_2 & v_2 \end{vmatrix} \mathbf{k},$$

for $\mathbf{i} = \langle 1,\, 0,\, 0 \rangle$, $\mathbf{j} = \langle 0,\, 1,\, 0 \rangle$, and $\mathbf{k} = \langle 0,\, 0,\, 1 \rangle$.

- **Magnitude/Norm**

: $\|\mathbf{v}\| \equiv \sqrt{{v_1}^2 + {v_2}^2 + {v_3}^3}$.

## · Extra Information

### - Eigenvalues & Eigenvectors

: For a vector space $V$ s.t. $\dim V = n \in \mathbb{N}$ and an $n \times n$ matrix $A$, we call each a scalar $\lambda$ and a nonzero vector $\mathbf{v} \in V$ an eigenvector and an eigenvalue of $A$, if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

### - Quadratic Surface *(From §11.6)*

: A surface able to express with

$$\left( \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} A & D & E \\ D & B & F \\ E & F & C \end{bmatrix} + \begin{bmatrix} G & H & I \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ z \end{bmatrix} + J = 0$$

or

$$\sum_{\substack{(i,\,j,\,k)\,\in\,P \\ 1 \leqslant n \leqslant 10}} K_n x^i y^j z^k$$

for $P = \{(i,\, j,\, k) \in \mathbb{N}_0^3 : i + j + k \leqslant 2\}$.

## · Inner/Dot Product

### - Definition

: Inner product of $\mathbf{u}$ and $\mathbf{v} \in V_n$ is defined to a function

$$\cdot : V_n \times V_n \longrightarrow \mathbb{R},$$

where $\mathbf{u} \cdot \mathbf{v} \equiv \sum_{i=1}^{n} u_i v_i$,

for $\mathbf{u} = \langle u_1,\, u_2,\, \ldots,\, u_n \rangle$ and $\mathbf{v} = \langle v_1,\, v_2,\, \ldots,\, v_n \rangle$.

### - Theorems ($\mathbf{u},\, \mathbf{v},\, \mathbf{w} \in V_n,\, c \in \mathbb{R}$)

(i) $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$.
(ii) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
(iii) $\mathbf{v} \cdot (\mathbf{u} + \mathbf{w}) = \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{w}$.
(iv) $(c\mathbf{u}) \cdot v = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$.
(v) $\mathbf{0} \cdot \mathbf{v} = 0$.

### - Angle Between Vectors

· For $\theta = \angle(\mathbf{u},\, \mathbf{v})$, $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$, i.e.,

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

· $\mathbf{u} \perp \mathbf{v}$ iff $\mathbf{u} \cdot \mathbf{v} = 0$. ($\because \cos 90^\circ = 0$)

### - Direction Cosines

· For $\begin{cases} \alpha = \angle(\overrightarrow{x_+},\, \mathbf{v}), \\ \beta = \angle(\overrightarrow{y_+},\, \mathbf{v}), \\ \gamma = \angle(\overrightarrow{z_+},\, \mathbf{v}), \end{cases}$ we say $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ the direction cosines of $\mathbf{v}$.

· $\cos\alpha = \dfrac{v_1}{\|\mathbf{v}\|}$, $\cos\beta = \dfrac{v_2}{\|\mathbf{v}\|}$, $\cos\gamma = \dfrac{v_3}{\|\mathbf{v}\|}$.

· $\dfrac{1}{\|\mathbf{v}\|}\mathbf{v} = (\text{unit vector of } \mathbf{v}) = \langle \cos\alpha,\, \cos\beta,\, \cos\gamma \rangle$.

### - Vector/Scalar Projection

· $\operatorname{comp}_{\mathbf{u}} \mathbf{v} \equiv \|\mathbf{v}\| \cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|}$.

· $\operatorname{proj}_{\mathbf{u}} \mathbf{v} \equiv (\operatorname{comp}_{\mathbf{u}} \mathbf{v}) \dfrac{\mathbf{u}}{\|\mathbf{u}\|} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2}\mathbf{u}$.

### - Work

: $W\,[\mathrm{J}] = \mathbf{F} \cdot \mathbf{D}\,[\mathrm{N{\cdot}m}]$.

### - Extra Information: Regression Analysis

: For $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$,
find $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ minimizing $\boldsymbol{\epsilon} = \mathbf{y} - X\boldsymbol{\beta}$.

$$\begin{aligned} \|\mathbf{y} = X\boldsymbol{\beta}\|^2 &= \|\mathbf{y}\|^2 + \|X\boldsymbol{\beta}\|^2 - 2\mathbf{y} \cdot X\boldsymbol{\beta} \\ &= \mathbf{y}^\top\mathbf{y} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta} - 2\mathbf{y}^\top X\boldsymbol{\beta}, \\ &\xrightarrow{\partial/\partial\boldsymbol{\beta}} \boldsymbol{\beta}^\top X^\top X - 2\mathbf{y}^\top X = \mathbf{0};\ \mathbf{y} = \frac{1}{2}X\boldsymbol{\beta}. \end{aligned}$$

## · Vector/Outer/Cross Product

### - Determinants

· $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \equiv ad - bc.$

· $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \equiv \sum_{k=1}^{3} (-1)^{1+k} a_{1k} \det(A_{1k}).$

*(For more info, go to LnAlgebra I.)*

### - Definition

$$\begin{aligned} : \mathbf{u} \times \mathbf{v} &\equiv \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \\ &= \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}. \end{aligned}$$

### - Properties & Theorems

· $\mathbf{u} \perp (\mathbf{u} \times \mathbf{v})$, $\mathbf{v} \perp (\mathbf{u} \times \mathbf{v})$.

· $\mathbf{u} \parallel \mathbf{v}$ iff $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.

· $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.

### - Triple Products

· Scalar triple product

$$\begin{aligned} : \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= \begin{vmatrix} \mathbf{a} \\ \mathbf{b} \\ \mathbf{c} \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\ &= (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}. \end{aligned}$$

· Vector triple product

: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$.

## · Equation of Lines/Planes

### - Line: Vector Equation

: For $\mathbf{r}_0 = \overrightarrow{\mathrm{OP}_0}$ and $\mathbf{v} \in \mathbb{R}^3$, an equation of a line parallel to $\mathbf{v}$ and containing $\mathrm{P}_0$ is

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}.$$

### - Line: Parametic Equation

: An equation

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

shows a line $/\!/ \langle a, b, c\rangle$ and contains $(x_0, y_0, z_0)$.

### - Line: Symmetric Equation

: For $\mathbf{v} = \langle a, b, c\rangle$ and $\overrightarrow{\mathrm{OP}_0} = \langle x_0, y_0, z_0\rangle$,

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

represents a line $\mathbf{r} = \overrightarrow{\mathrm{OP}_0} + t\mathbf{v}$.

### - Line: Vector Equation of segment

: For $\mathbf{r}_0$, $\mathbf{r}_1 \in \mathbb{R}^3$, an equation

$$\mathbf{r} = (1-t)\mathbf{r}_0 + t\mathbf{r}_1 \quad (0 \leqslant t \leqslant 1)$$

represents a segment from $\mathbf{r}_0$ to $\mathbf{r}_1$.

### - Plane: Vector Equation

: An equation

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

represents a plane such that

$$\begin{cases} \text{(i) its normal vector is } \mathbf{n}, \\ \text{(ii) it contains } \mathrm{P}_0 \text{ s.t. } \mathbf{r}_0 = \overrightarrow{\mathrm{OP}_0}. \end{cases}$$

### - Plane: Scalar Equation

: An equation

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

represents a plane such that

$$\begin{cases} \text{(i) its normal vector is } \langle a, b, c\rangle, \\ \text{(ii) it contains } \mathrm{P}_0(x_0, y_0, z_0). \end{cases}$$

### - Plane: Containing 3 Points

: An equation

$$(\overrightarrow{\mathrm{PQ}} \times \overrightarrow{\mathrm{PR}}) \cdot (\mathbf{r} - \overrightarrow{\mathrm{OP}}) = 0$$

represents a plane containing 3 points P, Q, R.

### - Intersection Point between Plane & Line

: For $\begin{cases} x = x_0 + at, \ y = y_0 + bt, \ z = z_0 + ct, \\ px + qy + rz = s, \end{cases}$

find $t$ such that

$$p(x_0 + at) + q(y_0 + bt) + r(z_0 + ct) = s.$$

### - Angle between 2 Planes

: $\theta = \cos^{-1}\left(\dfrac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\|\|\mathbf{n}_2\|}\right)$.

(from $\mathbf{n}_1 \cdot \mathbf{n}_2 = \|\mathbf{n}_1\|\|\mathbf{n}_2\| \cos\theta$)

### - Intersection Line between 2 Planes

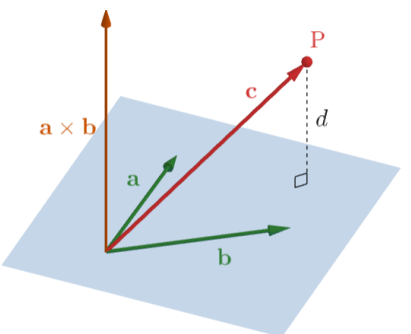
: For normal vectors $\mathbf{n}_1$ & $\mathbf{n}_2$, set $\mathbf{v} = \mathbf{n}_1 \times \mathbf{n}_2$.

### - Distance between Point and Plane

: The distance between a plane $ax + by + cz + d = 0$, and a point $(x_1, y_1, z_1)$ is

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

· $d = \left|\mathrm{comp}_{\mathbf{a}\times\mathbf{b}}\,\mathbf{c}\right| = \dfrac{|(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}|}{\|\mathbf{a}\times\mathbf{b}\|}$.



# §11. Vec.'s & Geometry of Space

## · Vector Functions & Space Curve

### - Definition: Vector Functions

: For a v.space $V$, a vector function is defined by a map

$$\mathbf{r}\colon \mathbb{R} \longrightarrow V.$$

Especially, in Calculus, for real-valued function $f$, $g$ and $h$, a map

$$\mathbf{r}\colon \mathbb{R} \longrightarrow \mathbb{R}^3 (= V_3)$$

s.t. $\mathbf{r}(t) = \langle f(t),\, g(t),\, h(t)\rangle$ is called a vector function.

### - Limit of Vector Function

$$: \lim_{t\to a} \mathbf{r}(t) \equiv \left\langle \lim_{t\to a} f(t),\, \lim_{t\to a} g(t),\, \lim_{t\to a} h(t) \right\rangle.$$

### - Space Curve

: A space curve defined on $I$ is a set

$$\{(f(t),\, g(t),\, h(t) : t \in I\}$$

for real-valued functions $f$, $g$ and $h$.

## · Calculus of Vector Functions

### - Derivative of Vector Function

$$: \mathbf{r}'(t) = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \equiv \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}.$$

### - If $\mathbf{r}(t) = \langle f(t),\, g(t),\, h(t)\rangle$ and $f$, $g$ and $h$ are d.able,

$$\mathbf{r}'(t) = \langle f'(t),\, g'(t),\, h'(t)\rangle.$$

### - Unit Tangent Vector $\mathbf{T}$

$$: \mathbf{T}(t) \equiv \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}.$$

### - Differentiation Rules

($\mathbf{u}$, $\mathbf{v}$: vector fcn., $f$: real-valued fcn., $c$: const.)

- $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}+\mathbf{v}) = \mathbf{u}' + \mathbf{v}'.$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}(c\mathbf{v}) = c\mathbf{v}'.$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}(f\mathbf{v}) = f'\mathbf{v} + f\mathbf{v}'.$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}\cdot\mathbf{v}) = \mathbf{u}'\cdot\mathbf{v} + \mathbf{u}\cdot\mathbf{v}'.$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}\times\mathbf{v}) = \mathbf{u}'\times\mathbf{v} + \mathbf{u}\times\mathbf{v}'.$
- $\dfrac{\mathrm{d}}{\mathrm{d}t}(\mathbf{u}\circ f) = f'(\mathbf{u}'\circ f).$ (Chain Rule)

### - If $\|\mathbf{r}(t)\|$ is const., then $\mathbf{r} \perp \mathbf{r}'$ for all $t$.

($\|\mathbf{r}(t)\| = \sqrt{f^2+g^2+h^2}$: const. $\iff$ $\mathbf{r}$ is on a sphere.)

### - $\mathbf{T}(t) \perp \mathbf{T}'(t)$ for all $\mathbf{T}$, $t$.

### - Integrals of Vector Function

#### · Definite Integrals

$$: \int_a^b \mathbf{r}(t)\,\mathrm{d}t \equiv \left\langle \int_a^b f(t)\,\mathrm{d}t,\, \int_a^b g(t)\,\mathrm{d}t,\, \int_a^b h(t)\,\mathrm{d}t \right\rangle.$$

#### · Antiderivatives

$$: \int \mathbf{r}(t) \equiv \left\langle \int f(t)\,\mathrm{d}t,\, \int g(t)\,\mathrm{d}t,\, \int h(t)\,\mathrm{d}t \right\rangle + \mathbf{C}.$$

($\mathbf{C}$: constant vector.)

## · Length of Arcs

### - Constant Length

$$: L = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\,\mathrm{d}t$$
$$= \int_a^b \|\mathbf{r}'(t)\|\,\mathrm{d}t.$$

### - Even if we change the parameter $t$ to sth. else, i.e., parametrization, the value of $L$ does not change.

### - Arc Length Function

$$: s(t) \equiv \int_a^t \|\mathbf{r}'(u)\|\,\mathrm{d}u$$
$$= \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}u}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}u}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}u}\right)^2}\,\mathrm{d}u.$$

## · Curvative

### - Definition

$$: \kappa(t) \equiv \left\|\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}s}\right\|.$$

### - $\kappa = \dfrac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \dfrac{\|\mathbf{r}'(t)\times\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}.$

pf.

We know that $\mathbf{T} = \dfrac{\mathbf{r}'}{\|\mathbf{r}'\|}$, and $\|\mathbf{r}'\| = \dfrac{\mathrm{d}s}{\mathrm{d}t}$,

$$\therefore\ \mathbf{r}' = \|\mathbf{r}'\|\mathbf{T} = \frac{\mathrm{d}s}{\mathrm{d}t}\mathbf{T},\quad \mathbf{r}'' = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2}\mathbf{T} + \frac{\mathrm{d}s}{\mathrm{d}t}\mathbf{T}'.$$

$$\implies \mathbf{r}'\times\mathbf{r}'' = \frac{\mathrm{d}s}{\mathrm{d}t}\frac{\mathrm{d}^2 s}{\mathrm{d}t^2}\cancelto{0}{(\mathbf{T}\times\mathbf{T})} + \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2(\mathbf{T}\times\mathbf{T}'),$$

$$\implies \|\mathbf{r}'\times\mathbf{r}''\| = \left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 \|\mathbf{T}\|\|\mathbf{T}'\| = \|\mathbf{r}'\|^2\|\mathbf{T}'\|,$$

since $\mathbf{T}\perp\mathbf{T}'$, and $\|\mathbf{T}\| = 1$.

$$\implies \frac{\|\mathbf{r}'\times\mathbf{r}''\|}{\|\mathbf{r}'\|^3} = \frac{\|\mathbf{T}'\|}{\|\mathbf{r}'\|} = \kappa.$$ ■

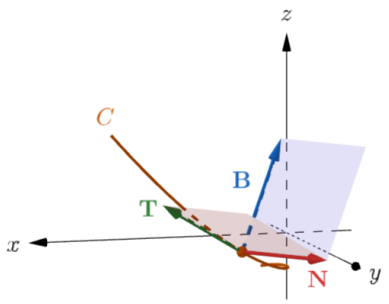
## · Normal Vectors

- **N: (Principal) Unit Normal Vector**
  : $\mathbf{N}(t) \equiv \dfrac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$.

- **B: Binormal Vector**
  : $\mathbf{B}(t) \equiv \mathbf{T}(t) \times \mathbf{N}(t)$.

- Each $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ is a unit vector, and perpendicular with each other.

- **Planes**
  · Normal Plane: A plane made by $\mathbf{N}$ and $\mathbf{B}$.
  · Osculting Plane: A plane made by $\mathbf{N}$ and $\mathbf{T}$.




## · Velocity & Acceletation

- **Definition**
  · **Velocity**: $\mathbf{v}(t) \equiv \lim_{h\to 0} \dfrac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h} = \mathbf{r}'(t)$.
  · **Speed**: $v = \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|$.
  · **Acceleration**: $\mathbf{a}(t) \equiv \mathbf{v}'(t) = \mathbf{r}''(t)$.

- **Separating Acceleration to Tangent/Normal**
  : For position vector $\mathbf{r}(t)$, let $v = \|\mathbf{v}(t)\|$. Then

$$\mathbf{a} = v'\mathbf{T} + v^2\kappa\mathbf{N}.$$

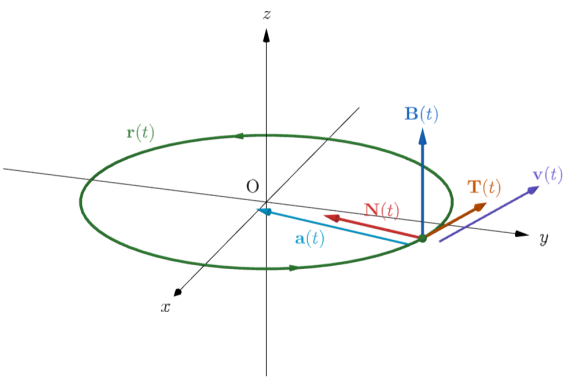
pf.

$$\begin{aligned}\mathbf{a}(t) &= \frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathbf{r}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}(t)\right)\\ &= \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{r}'(t) = \frac{\mathrm{d}}{\mathrm{d}t}[v\mathbf{T}(t)]\\ &= v'\mathbf{T}(t) + v\mathbf{T}'(t)\\ &= v'\mathbf{T}(t) + v\|\mathbf{T}'(t)\|\mathbf{N}(t)\\ &= v'\mathbf{T}(t) + v^2\kappa\mathbf{N}(t),\end{aligned}$$

$\therefore\ \mathbf{a} = v'\mathbf{T} + v^2\kappa\mathbf{N}.$ ■

- **Newton's laws of motion**
  : $\mathbf{F} = m\mathbf{a}$.

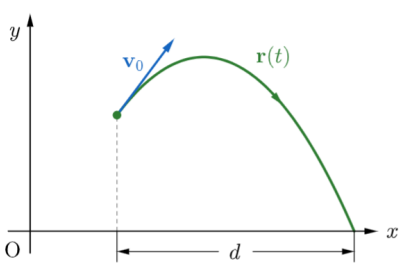
- **Example: Circular Works** $(a > 0)$
  · $\mathbf{r}(t) = \langle a\cos t,\ a\sin t,\ 0\rangle$.
  · $\mathbf{v}(t) = \langle -a\sin t,\ a\cos t,\ 0\rangle$.
  $\implies v(t) = \|\mathbf{v}(t)\| = a$.
  · $\mathbf{a}(t) = \langle -a\cos t,\ -a\sin t,\ 0\rangle$.
  · $\mathbf{T}(t) = \langle -\sin t,\ \cos t,\ 0\rangle$.
  $\implies \|\mathbf{T}(t)\| = 1$.
  · $\kappa = \dfrac{1}{a}$.
  · $\mathbf{T}'(t) = \langle -\cos t,\ -\sin t,\ 0\rangle$.
  $\implies \|\mathbf{T}'(t)\| = 1$.
  · $\mathbf{N}(t) = \langle -\cos t,\ -\sin t,\ 0\rangle\ (= \mathbf{r}(t)/a)$.
  · $\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \langle 0,\ 0,\ 1\rangle$.




- **Example: Projectiles** $(0 < \alpha < \pi/2,\ \|\mathbf{v}_0\| = v_0 > 0)$
  · $\mathbf{a}(t) = -g\,\mathbf{j}$. ($g = 9.8\,\mathrm{m/s^2}$: Gravity constant)
  · $\mathbf{v}(t) = \displaystyle\int_0^t \mathbf{a}(u)\,\mathrm{d}u + \mathbf{v}_0 = \mathbf{v}_0 - g\,\mathbf{j}$.
  · $\mathbf{v}_0 = v_0\cos\alpha\,\mathbf{i} + v_0\sin\alpha\,\mathbf{j}$.

$$\begin{aligned}\implies \mathbf{r}(t) &= \int_0^t \mathbf{v}(u)\,\mathrm{d}u + \mathbf{r}_0 = t\mathbf{v}_0 - \frac{1}{2}gt^2\,\mathbf{j} + \mathbf{r}_0\\ &= (v_0 t\cos\alpha + x_0)\,\mathbf{i} + \left(v_0 t\sin\alpha - \frac{1}{2}gt^2 + y_0\right)\mathbf{j}.\end{aligned}$$

$\implies$ The eq. $\mathrm{proj}_{\mathbf{j}}\,\mathbf{r}(t) = 0$: Extract $t = t_0\ (> 0)$.
then, $d = \mathrm{proj}_{\mathbf{i}}\,\mathbf{r}(t_0) - \mathrm{proj}_{\mathbf{i}}\,\mathbf{r}(0)$.



# §13. Partial Derivatives

## · Functions of Many Variables

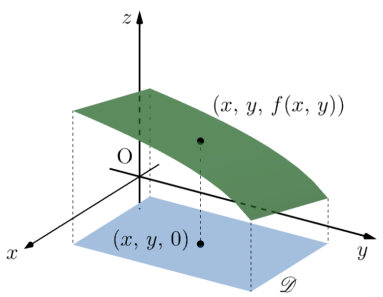
### - Definition

: $f\colon \begin{array}{ccc} D(\subseteq \mathbb{R}^n) & \longrightarrow & \mathbb{R} \\ (x_1, x_2, \dots, x_n) & \longmapsto & y = f(x_1, x_2, \dots, x_n) \end{array}$
is a function of many variables.

In this book, we usually treat
$f_2\colon \mathbb{R}^2 \longrightarrow \mathbb{R}$ and $f_3\colon \mathbb{R}^3 \longrightarrow \mathbb{R}$.

### - Graph of $z = f(x, y)$

: Graph of $f\colon D(\subseteq \mathbb{R}^2) \longrightarrow \mathbb{R}$ is defined by a set
$$\{(x, y, f(x, y)) : (x, y) \in D\}.$$




If $z = f(x, y)$ is 'linear equation,' $(z = ax + by + c)$
then its graph expresses a plane.

## · Limit & Continuity

### - Limit

: $\lim\limits_{(x,\,y)\to(a,\,b)} f(x, y) = L$, if, given $\epsilon > 0$, $\exists \delta > 0$ s.t.
$$\sqrt{(x-a)^2 + (y-b)^2} < \delta \ \text{ implies } \ |f(x) - L| < \epsilon.$$

### - Operation of Limits

: Let $\mathrm{li} = \lim\limits_{(x,\,y)\to(a,\,b)}$. If each limit exists, then

· $\mathrm{li}\,(f \pm g) = \mathrm{li}\, f \pm \mathrm{li}\, g$.

· $\mathrm{li}\,(cf) = c\,\mathrm{li}\, f$.

· $\mathrm{li}\,(fg) = \mathrm{li}\, f\, \mathrm{li}\, g$.

· $\mathrm{li}\left(\dfrac{f}{g}\right) = \dfrac{\mathrm{li}\, f}{\mathrm{li}\, g}$. $(\mathrm{li}\, g \neq 0)$

### - Continuity

· $f\colon \mathbb{R}^2 \longrightarrow \mathbb{R}$ is continuous at $(a, b)$, if
$\lim\limits_{(x,\,y)\to(a,\,b)} f(x, y) = f(a, b)$.

· $f$ is cont. on $D$, if $f$ is cont. at every $(a, b) \in D$.

## · Partial Derivative

### - Definition

· $f_x(x, y) = \dfrac{\partial f}{\partial x} \equiv \lim\limits_{h\to 0} \dfrac{f(x+h, y) - f(x, y)}{h}$.

· $f_y(x, y) = \dfrac{\partial f}{\partial y} \equiv \lim\limits_{h\to 0} \dfrac{f(x, y+h) - f(x, y)}{h}$.

### - Higher Partial Derivative

· $\dfrac{\partial^2 f}{\partial x^2} \equiv \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial x}\right)$.

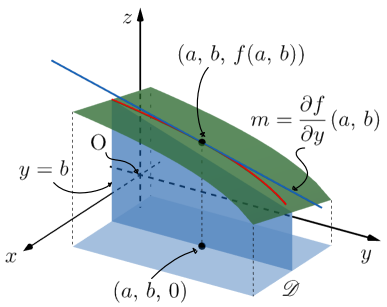
· $\dfrac{\partial^2 f}{\partial y^2} \equiv \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial y}\right)$.

· $\dfrac{\partial^2 f}{\partial x\,\partial y} \equiv \dfrac{\partial}{\partial x}\left(\dfrac{\partial f}{\partial y}\right)$.

· $\dfrac{\partial^2 f}{\partial y\,\partial x} \equiv \dfrac{\partial}{\partial y}\left(\dfrac{\partial f}{\partial x}\right)$.

### - Meaning on Geometry

: $f_y(a, b)$ is the slope of tangent line of $z = f(x, y)$,
in plane $y = b$.




### - Many Variables

: For $f\colon \prod\limits_{i=1}^{n}[a_i, b_i](\subseteq \mathbb{R}^n) \longrightarrow \mathbb{R}$,

$$\frac{\partial f}{\partial x_i} \equiv \lim_{h\to 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(\dots, x_i, \dots)}{h}.$$

### - Clairaut's Theorem

: If both $\partial f/(\partial x\,\partial y)$ and $\partial f/(\partial y\,\partial x)$ are cont. on $D$,
then $\dfrac{\partial f}{\partial x\,\partial y} = \dfrac{\partial f}{\partial y\,\partial x}$.

### - Partial Differential Equation

#### · Laplace's Equation

: Find the harmony ftn. $u$ s.t. $\Delta u \equiv \sum\limits_{i=1}^{n} \dfrac{\partial^2 u}{\partial {x_i}^2} = 0$.

ex) $e^x \sin y$

#### · Wave Equation

: Find a ftn. $u$ s.t. $\dfrac{\partial^2 u}{\partial t^2} = a^2 \dfrac{\partial^2 u}{\partial x^2}$ for $a \in \mathbb{R}$.

ex) $\sin(x - at)$

## · Tangent Plane & Linear Approximation

### - Tangent Plane

: The eq. of the tan. plane of $z = f(x, y)$ at $(x_0, y_0)$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

### - Linear Approximation

: For $A = f_x(x_0, y_0)$ and $B = f_y(x_0, y_0)$,

$f(x, y) \approx f(x_0, y_0) + A(x - x_0) + B(y - y_0)$.

### - Differentiable

#### · Definition

: For $z = f(x, y)$, $f$ is diff.ble at $(a, b)$, if $\exists \epsilon_1, \epsilon_2$ s.t. $(\epsilon_1, \epsilon_2) \longrightarrow (0, 0)$ as $(\Delta x, \Delta y) \longrightarrow (0, 0)$, and

$$\Delta z = (f_x(a, b) + \epsilon_1)\Delta x + (f_y(a, b) + \epsilon_2)\Delta y.$$

#### · Judging Theorem of Diff.ble

: If $f_x$ and $f_y$ exists & cont. at $(a, b)$, then $f$ is differentiable at $(a, b)$.

pf.

If $f$ is diff.ble at $x = a$,
then $\dfrac{\Delta y}{\Delta x} \longrightarrow f'(a)$ as $\Delta x \longrightarrow 0$,
thus if we define $\epsilon = \dfrac{\Delta y}{\Delta x} - f'(a)$, then $\lim\limits_{\Delta x \to 0} \epsilon = 0$,
and we can express $\Delta y = f'(a)\Delta x + \epsilon \Delta x$.

Then, for $f(x, y)$, we can define

$$\epsilon_1' = \frac{f(a+\Delta x, b+\Delta y) - f(a, b+\Delta y)}{\Delta x} - f_x(a, b+\Delta y),$$

$$\epsilon_2 = \frac{f(a, b+\Delta y) - f(a, b)}{\Delta y} - f_y(a, b),$$

where $(\epsilon_1', \epsilon_2) \longrightarrow (0, 0)$ as $(\Delta x, \Delta y) \longrightarrow (0, 0)$,
and

$$\begin{aligned}\Delta z &= f(a + \Delta x, b + \Delta y) - f(a, b) \\ &= f(a + \Delta x, b + \Delta y) - f(a, b + \Delta y) \\ &\quad + f(a, b + \Delta y) - f(a, b) \\ &= \Delta f_{\Delta x} + \Delta f_{\Delta y} \\ &= (f_x(a, b + \Delta y) + \epsilon_1')\Delta x + (f_y(a, b) + \epsilon_2)\Delta y.\end{aligned}$$

But, if $f_x$ is cont. at $(a, b)$, then

$$\lim_{\Delta y \to 0} f_x(a, b + \Delta y) = f_x(a, b),$$

thus if we define $\epsilon_1'' = f_x(a, b + \Delta y) - f_x(a, b)$,
then $\lim\limits_{\Delta y \to 0} \epsilon_1'' = 0$.

Then we have

$$\begin{aligned}\Delta z &= (f_x(a, b) + \epsilon_1'' + \epsilon_1')\Delta x + (f_y(a, b) + \epsilon_2)\Delta y \\ &= (f_x(a, b) + \epsilon_1)\Delta x + (f_y(a, b) + \epsilon_2)\Delta y\end{aligned}$$

for $\epsilon_1 = \epsilon_1' + \epsilon_1''$, where

$$\lim_{(\Delta x, \Delta y) \to (0, 0)} \epsilon_1 = \lim_{\Delta x \to 0} \epsilon_1' + \lim_{\Delta y \to 0} \epsilon_1'' = 0 + 0 = 0.$$

■

### - Total Derivative/Differential

: $\mathrm{d}z \equiv \dfrac{\partial z}{\partial x}\,\mathrm{d}x + \dfrac{\partial z}{\partial y}\,\mathrm{d}y$. $\left(= \lim\limits_{(\Delta x, \Delta y) \to (0, 0)} \Delta z\right)$

## · Chain Rule

### - Case 1: $z = f(x(t), y(t))$

: $\dfrac{\mathrm{d}z}{\mathrm{d}t} = \dfrac{\partial z}{\partial x}\dfrac{\mathrm{d}x}{\mathrm{d}t} + \dfrac{\partial z}{\partial y}\dfrac{\mathrm{d}y}{\mathrm{d}t}$.

### - Case 2: $z = f(x(s, t), y(s, t))$

: $\dfrac{\partial z}{\partial s} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial s}$, $\dfrac{\partial z}{\partial t} = \dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial t} + \dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial t}$.

### - General Case

: If $z = f(x_1, x_2, \ldots, x_n)$ with each $x_j(t_1, t_2, \ldots, t_m)$,

$$\frac{\partial z}{\partial t_i} = \sum_{j=1}^{n} \frac{\partial z}{\partial x_j}\frac{\partial x_j}{\partial t_i}.$$

- **Implicit Funtion Theorem**

  : For $F(x, y) = 0$ where $y = f(x)$, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{F_x}{F_y}$. $(F_y \neq 0)$

  pf.

  Since $y$ is ftn. of $x$, we have

  $$\frac{\partial F}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}x} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x} = 0,$$

  thus, if $\partial F/\partial y \neq 0$,

  $$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\dfrac{\partial F}{\partial x}}{\dfrac{\partial F}{\partial y}} = -\frac{F_x}{F_y}.$$

  ■

## · Directional Derivatives & Gradient Vectors

- **Directional Derivatives**

  · **Definition**

  : The directional drv. of $f$ to $\mathbf{u}$ direction is def. by

  $$D_{\mathbf{u}}f(x_0, y_0) \equiv \lim_{h\to 0}\frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

  for $\mathbf{u} = \langle a, b\rangle$.

  · $D_{\mathbf{u}}f = a\dfrac{\partial f}{\partial x} + b\dfrac{\partial f}{\partial y}$.

  pf.

  Let $x_0' = x_0 + ha$, $y_0' = y_0 + hb$, $\mathbf{x}' = (x_0', y_0')$, $\mathbf{x} = (x_0, y_0)$.

  $$\begin{aligned} D_{\mathbf{u}}f(\mathbf{x}) &= \lim_{h\to 0}\frac{f(\mathbf{x}') - f(\mathbf{x})}{h} \\ &= \lim_{h\to 0}\frac{1}{h}\begin{pmatrix} f(\mathbf{x}') - f(x_0, y_0') \\ +f(x_0, y_0') - f(\mathbf{x}) \end{pmatrix} \\ &= \lim_{h\to 0}\begin{pmatrix} a(f(\mathbf{x}') - f(x_0, y_0'))/(ah) \\ +b(f(x_0, y_0') - f(\mathbf{x}))/(bh) \end{pmatrix} \\ &= a\frac{\partial f}{\partial x} + b\frac{\partial f}{\partial y}. \end{aligned}$$

  ■

- **Gradient Vectors**

  · **Definition**

  : The gradient vector of $f$ at $(x_0, y_0)$ is defined by

  $$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right\rangle \quad \text{or} \quad \nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right\rangle.$$

  · $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

  · **Maximum of $D_{\mathbf{u}}f$**

  : $D_{\mathbf{u}}f \leqslant \|\nabla f\|$. $(D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \|\nabla f\|\|\mathbf{u}\|\cos\theta)$

  "=" stands if $\mathbf{u} \parallel \nabla f$.

- **Tangent Plane to Level Surfaces**

  : The equation of tangent plane to the level surface $f(x, y, z) = k$, at $\mathrm{P}(x_0, y_0, z_0)$ is

  $$\nabla f(\mathbf{r}_0) \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

  with $\mathbf{r} = \langle x, y, z\rangle$, $\mathbf{r}_0 = \overrightarrow{\mathrm{OP}}$.

## · Maximum & Minimum

- **Global Max/min.**

  · $f$ is (global) maximum at $(a, b)$, if $f(a, b) \geqslant f(x, y)$ for all $(x, y)$ in domain of $f$.

  · $f$ is (global) minimum at $(a, b)$, if $f(a, b) \leqslant f(x, y)$ for all $(x, y)$ in domain of $f$.

- **Local Max/min.**

  · $f$ is local maximum at $(a, b)$, if $f(a, b) \geqslant f(x, y)$ for all $(x, y)$: nbh. of $(a, b)$.

  · $f$ is local minimum at $(a, b)$, if $f(a, b) \leqslant f(x, y)$ for all $(x, y)$: nbh. of $(a, b)$.

- **First Derivatives Test**

  : If $f$ has local max/min. and $\exists$ pt.deriv.s at $(a, b)$, then $\nabla f(a, b) = \mathbf{0}$.

- **Critical Point**

  : $(a, b)$ is a critical point of $f$, if $\nexists$ any pt.deriv. or $f_{xy}f_{yx} = 0$ at $(a, b)$.

- **Second Derivative Test**

  : For $f(x, y)$ s.t. $f_{xy}$ & $f_{yx}$ are cont. and $= 0$ at $(a, b)$, let $D = \begin{vmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{yx}(a, b) & f_{yy}(a, b)\end{vmatrix}$. Then

  (i) If $D > 0$ & $f_{xx} > 0$, then $f$: local min. at $(a, b)$.

  (ii) If $D > 0$ & $f_{xx} < 0$, then $f$: local max. at $(a, b)$.

  (iii) If $D < 0$, then $f$ has a saddle point at $(a, b)$.

- **Concaveness**
  - **Definition**
    : For a function $f: D(\subseteq \mathbb{R}) \longrightarrow \mathbb{R}$
    and its tangent line $y = L_{x_0}(x)$ at $x_0 \in I \subseteq D$,
    - $f$ is concave upward(CU), if for each $x_0 \in I$, $f(x) - L_{x_0}(x) > 0$ for all $x \in I \setminus \{x_0\}$.
    - $f$ is concave downward(CD), if for each $x_0 \in I$, $f(x) - L_{x_0}(x) < 0$ for all $x \in I \setminus \{x_0\}$.
  - If $f''(x) > 0$ on $I$, then $f$ is CU on $I$.
  - If $f''(x) < 0$ on $I$, then $f$ is CD on $I$.

pf. (CU)

Since $L_{x_0}(x) = f(x_0) + f'(x_0)(x - x_0)$,
it suffices to check

$$f(x) > f(x_0) + f'(x_0)(x - x_0).$$

for $x \in I \setminus \{x_0\}$.

Since $f''(x) > 0$, $f'$ is increasing on $I$, thus
if $x > x_0$, then $f'(x) > f'(x_0)$ on $\{x \in I : x > x_0\}$,
if $x < x_0$, then $f'(x) < f'(x_0)$ on $\{x \in I : x < x_0\}$.

Also, by MVT, $\exists c$ between $x_0$ and $x$ s.t.

$$\frac{f(x) - f(x_0)}{x - x_0} = f'(c).$$

Then we have if $x > x_0$, then $f'(c) > f'(x_0)$,
thus $\dfrac{f(x) - f(x_0)}{x - x_0} > f'(x_0)$, hence we conclude
$f(x) > f(x_0) + f'(x_0)(x - x_0)$, since $x - x_0 > 0$.

On the other side, if $x < 0$, then $f'(c) < f'(x_0)$,
so $\dfrac{f(x) - f(x_0)}{x - x_0} > f'(x_0)$, therefore we take
$f(x) > f(x_0) + f'(x_0)(x - x_0)$, since $x - x_0 < 0$. ■

- **Lagrange Multiplier Method**
  : Critical points of $f(\mathbf{x})$ subject to $g(\mathbf{x}) = k$ can be found by solving following eq. of $x$, $y$, $z$ and $\lambda$:

$$\begin{cases} \nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x}), \\ g(\mathbf{x}) = k. \end{cases}$$

[$\mathbf{x}$ is a two- or three-dimension variable vector.]

# §14. Multiple Integrals

- **Double Integrals** (over rectangle)
  - **Definition**
    : The double integral of $f$ over $R$ is defined by

$$\iint_R f(x, y)\,\mathrm{d}A \equiv \lim_{\substack{m\to\infty \\ n\to\infty}} \sum_{i=1}^{m}\sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*)\Delta A.$$

(Sets of $\Delta A$, $x_{ij}^*$, etc. are sim. to ones of single int.)

  - For nonnegative function $f: R(\subseteq \mathbb{R}^2) \longrightarrow \mathbb{R}$,

$$V = \iint_R f(x, y)\,\mathrm{d}A.$$

  - **Fubini's Theorem**
    : For $R = [a, b] \times [c, d]$ and $f: R \longrightarrow \mathbb{R}$,

$$\begin{aligned} \iint_R f(x, y)\,\mathrm{d}A &= \int_a^b \int_c^d f(x, y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_c^d \int_a^b f(x, y)\,\mathrm{d}x\,\mathrm{d}y \end{aligned}$$

    if $f$ is cont. on $R$ (or bdd. & finitely discont. on $R$).

  - For a seperatable function $f$ on $R = [a, b] \times [c, d]$,
    i.e., $f(x, y) = g(x)h(y)$ for some $g$ and $h$,

$$\iint_R f(x, y)\,\mathrm{d}A = \int_a^b g(x)\,\mathrm{d}x \int_c^d h(y)\,\mathrm{d}y.$$

- **Double Integrals over General Regions**
  - **Type I**: Fixed $x$
    : If $D = \{(x, y) : a \leqslant x \leqslant b,\ g_1(x) \leqslant y \leqslant g_2(x)\}$,

$$\iint_D f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

  - **Type II**: Fixed $y$
    : If $D = \{(x, y) : h_1(y) \leqslant x \leqslant h_2(y),\ c \leqslant y \leqslant d\}$,

$$\iint_D f(x, y)\,\mathrm{d}A = \int_c^d \int_{h_1(x)}^{h_2(x)} f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

  - **General Region** $D$
    : $\displaystyle\iint_D f(x, y)\,\mathrm{d}A = \iint_R F(x, y)\,\mathrm{d}A$
    for some rectangular region $R$ s.t. $D \subseteq R$,
    and $F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D, \\ 0 & \text{if } (x, y) \in R \setminus D. \end{cases}$

## · Double Integrals in Polar Coord.

: For $R = \{(r, \theta) : 0 \leqslant a \leqslant r \leqslant b,\ \alpha \leqslant \theta \leqslant \beta\}$,

$$\iint_R f(x, y)\,\mathrm{d}A = \int_\alpha^\beta \int_a^b f(r\cos\theta, r\sin\theta)\, r\,\mathrm{d}r\,\mathrm{d}\theta.$$

pf. $(\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta)$

$$\begin{aligned}\Delta A &= \frac{1}{2}r_{i+1}^2\Delta\theta - \frac{1}{2}{r_i}^2\Delta\theta \\ &= \frac{r_{i+1}+r_i}{2}\Delta r\Delta\theta \\ &= \overline{r_i}\Delta r\Delta\theta \\ &\longrightarrow r\,\mathrm{d}r\,\mathrm{d}\theta \quad \text{as } \Delta r,\ \Delta\theta \longrightarrow 0,\end{aligned}$$

thus $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$. ■

## · Surface Area

: For a cont. ftn. $f : D(\subseteq \mathbb{R}^2) \longrightarrow \mathbb{R}$,
the area of surface $S$ on $D$ is

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}.$$

pf.

Write $\mathbf{r}(t) = \overrightarrow{(x_i, y_i, f(x_i, y_i))(x_{i+1}, y_i, f(x_{i+1}, y_i))}$
and $\mathbf{s}(t) = \overrightarrow{(x_i, y_i, f(x_i, y_i))(x_i, y_{i+1}, f(x_i, y_{i+1}))}$
with $(x_i, y_i) \in D$ and $x_{i+1} = x_i + t\Delta x$, $y_{i+1} = y_i + t\Delta y$.

Then each tangent vector of $\mathbf{r}$ and $\mathbf{s}$ is

$$\mathbf{r}'(t) = \Delta x\,\langle 1, 0, f_x\rangle, \quad \mathbf{s}'(t) = \Delta y\,\langle 0, 1, f_y\rangle$$

Then, $\Delta A = \|\mathbf{r}' \times \mathbf{s}'\| = \Delta x\Delta y\sqrt{{f_x}^2 + {f_y}^2 + 1}$. ■

## · Triple Integrals

### - Definition

: The triple integral of $f$ on a rectangle area $R$ is

$$\iiint_E f(\mathbf{x})\,\mathrm{d}V = \lim_{l,m,n\to\infty} \sum_{i=1}^{l}\sum_{j=1}^{m}\sum_{k=1}^{n} f(\mathbf{x}_{ijk}^*)\,\Delta V$$

with $\mathbf{x} = (x, y, z)$ and $\mathbf{x}_{ijk}^* = (x_{ijk}^*, y_{ijk}^*, z_{ijk}^*)$.

### - Fubini's Theorem on Triple Integrals

: If $f$ is continuous on $B = [a, b] \times [c, d] \times [r, s]$,

$$\iiint_B f(x, y, z)\,\mathrm{d}V = \int_r^s\int_c^d\int_a^b f(x, y, z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

($x$, $y$, $z$ can be shuffled; careful to range.)

### - Volume

: The volume of 3-dim area $V(E)$ is

$$V(E) = \iiint_E \mathrm{d}V.$$

### - On Cylinderical Coordinates

: Convert $(x, y, z)$ to $(r, \theta, z)$,
using $x = r\cos\theta$, $y = r\sin\theta$, $z = z$.

## · Examples of Calculating Integrals

- For $E$: on $(\mathbb{R}^+)^3$, bdd. with $z = 12xy$, $x = 1$, $y = x$,

$$\iiint_E z\,\mathrm{d}V$$

sol.

Given $x \in [0, 1]$, $0 \leqslant y \leqslant x$ and $0 \leqslant z \leqslant 12xy$.
Then we have

$$\begin{aligned}\iiint_E z\,\mathrm{d}V &= \int_0^1\int_0^x\int_0^{12xy} z\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1\int_0^x \frac{144}{2}x^2y^2\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 \frac{144}{6}x^5\,\mathrm{d}x = 4.\end{aligned}$$ ■

- For $E$: bounded with $y = x^2 + z^2$ and $y = 4$,

$$\iiint_E \sqrt{x^2+z^2}\,\mathrm{d}V$$

sol.

Given $y \in [0, 4]$, if we let $x = r\cos\theta$, $y = r\sin\theta$,
and $D = \{(x, z) : x^2 + z^2 \leqslant 4\}$,

$$\begin{aligned}\iiint_E \sqrt{x^2+z^2}\,\mathrm{d}V &= \int_0^{2\pi}\int_0^4\int_0^{\sqrt{y}} r\cdot r\,\mathrm{d}r\,\mathrm{d}\theta \\ &= 2\pi\int_0^4 \frac{y^{3/2}}{3}\,\mathrm{d}r \\ &= 2\pi\cdot\frac{1}{3}\cdot\frac{2}{5}y^{5/2}\Big]_0^4 = \frac{128}{15}\pi.\end{aligned}$$ ■

- Volume of bounded solid $E$ with
$x + 2y + z = 2$, $x = 2y$, $x = 0$, $z = 0$

sol.

Given $x \in [0, 1]$, $x/2 \leqslant y \leqslant 1 - x/2$,
$0 \leqslant z \leqslant 2 - x - 2y$. Then we have

$$\begin{aligned}\iiint_E \mathrm{d}V &= \int_0^1\int_{x/2}^{1-x/2}\int_0^{2-x-2y} \mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1\int_{x/2}^{1-x/2}(2 - x - 2y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^1 (x^2 - 2x + 1)\,\mathrm{d}x = \frac{1}{3}.\end{aligned}$$ ■

# §-. Continuous Random Variables

## · Denstity/Distribution Functions

- **Cumulative Distribution Function**
  : A function $F_X$ with $F_X(x) = \mathrm{P}(X \leqslant x)$.
  Note that every CDF are non-decreasing.

- **Probability Density Function**
  : A function $f\colon D(\subseteq \mathbb{R}) \longrightarrow \mathbb{R}$ s.t.
  · $f(x) \geqslant 0$ for all $x \in D$,
  · $\displaystyle\int_D f(x)\,\mathrm{d}x = 1$,
  · $\mathrm{P}(X \in A) = \displaystyle\int_A f(x)\,\mathrm{d}x$.

- For $x \in \mathbb{R}$ and an interval $A \subseteq \mathbb{R}$, note that
  $$\mathrm{P}(X = x) = 0, \text{ thus}$$
  $$\mathrm{P}(X \in A) = \mathrm{P}(X \in \overline{A}) = \mathrm{P}(X \in \mathring{A}).$$

- **Percentiles**
  : $(100p)$-th percentile (denoted by $\pi_p$) is defined by
  $$p = \int_{-\infty}^{\pi_p} f(x)\,\mathrm{d}x = \mathrm{P}(X \leqslant \pi_p) = F_X(\pi_p).$$
  · $\pi_{0.25} = q_1$: First quartile.
  · $\pi_{0.5} = q_2 = m$: Second quartile (Median).
  · $\pi_{0.75} = q_3$: Third quartile.

- **Two Normal Distribution**
  : For $X,\ Y \sim \mathrm{N}(\mu,\ \sigma^2)$, their (J)PDF is
  $$f(x,\ y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{(x-\mu)^2 + (y-\mu)^2}{2\sigma^2}\right).$$

  pf. $f$: PDF?

  Let $u = x - \mu,\ t = y - \mu$,
  then $u,\ t \in \mathbb{R},\ (\mathrm{d}u,\ \mathrm{d}t) = (\mathrm{d}x,\ \mathrm{d}y)$;
  $$\iint_{\mathbb{R}^2} f = \frac{1}{\sqrt{2\pi}\sigma}\int_{\mathbb{R}}\int_{\mathbb{R}} \exp\left(-\frac{u^2+t^2}{2\sigma^2}\right)\mathrm{d}u\,\mathrm{d}t,$$
  convert to polar coordinate:
  $$= \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{\infty} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\,\mathrm{d}r\,\mathrm{d}\theta$$
  $$= \frac{1}{\sigma^2}\int_0^{\infty} \exp\left(-\frac{r^2}{2\sigma^2}\right) r\,\mathrm{d}r$$
  $$= \exp\left(-\frac{r^2}{2\sigma^2}\right)\Big]_0^{\infty} = 0 - (-1) = 1.$$
  ■

  **cf.** For $X \sim \mathrm{N}(\mu,\ \sigma^2)$, its PDF is
  $$f(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

## · Expectations

- **Mean**
  : $\mu = \mathrm{E}(X) = \displaystyle\int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x.$

- **Variance**
  : $\mathrm{Var}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 \left(= \displaystyle\int_{\mathbb{R}} (x-\mu)^2 f(x)\,\mathrm{d}x\right).$

- **Standard Deviation**
  : $\sigma = \sqrt{\mathrm{Var}(X)}$.

- **Moment Gerenerating Function**
  : $M(t) = \displaystyle\int_{-\infty}^{\infty} e^{tx} f(x)\,\mathrm{d}x\ (= \mathrm{E}(e^{tX}))$.
  Note that $M'(0) = \mathrm{E}(X)$, $M''(0) - (M'(0))^2 = \mathrm{Var}(X)$.

## · Typical Distribution

- **Mixture Distribution**
  : For some PDF $f_1$, $f_2$ and $a \in [0,\ 1]$,
  $$f(x) = a f_1(x) + (1-a) f_2(x).$$

- **Piecewise Distribution**
  : For some ftn.s $f_1$ on $A$ and $f_2$ on $B$, $f$ on $A \cup B$ s.t.
  $$\int_{A\cup B} f = \int_A f_1 + \int_B f_2.$$

- **Uniform Distribution**
  · **Definition**
  : A distribution $U(a,\ b)$ s.t. for $X \sim U(a,\ b)$,
  $$f(x) = \frac{1}{b-a} \quad \text{for all } x \in (a,\ b).$$
  · **Expectations**
  - $\mathrm{E}(X) = \dfrac{a+b}{2}$.
  - $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$.
  - $M(t) = \dfrac{e^{tb} - e^{ta}}{t(b-a)}$.